

Name: Answers



Baldivis
Secondary College

Pythagoras Theorem Investigation Year 9

Time: 50 minutes

Marks: / 42

Marks: / 5

Allowed: Take home section of notes and a calculator

Question 1

10 marks (1/2 mark each)

A. Square the following numbers:

1	1
21	441
3	9
4	16
12	144
29	841
15	225
11	121
20	400
10	100

B. Square root the following numbers:

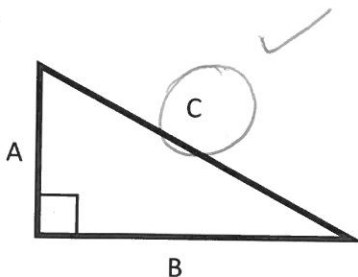
1	1
2	1.41
64	8
36	6
169	13
25	5
16	4
9	3
144	12
225	15

Question 2

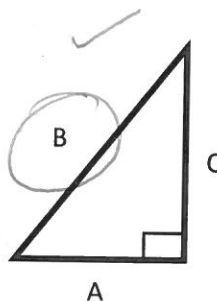
3 marks

Circle the hypotenuse on the following diagrams.

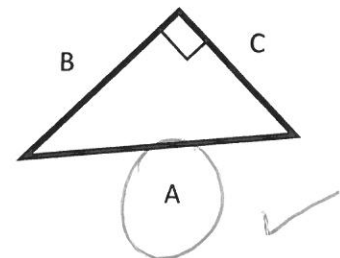
A.



B.



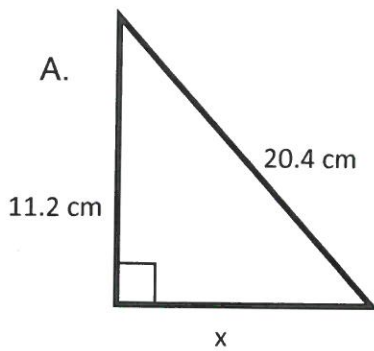
C.



Question 5

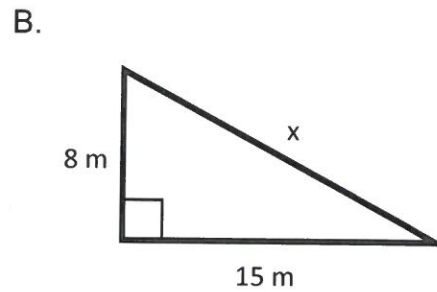
12 marks (2,2,4,4)

Calculate the unknown sides of the right angle-triangles. Show all working out.



$$20.4^2 - 11.2^2 = 290.72 \quad \checkmark$$

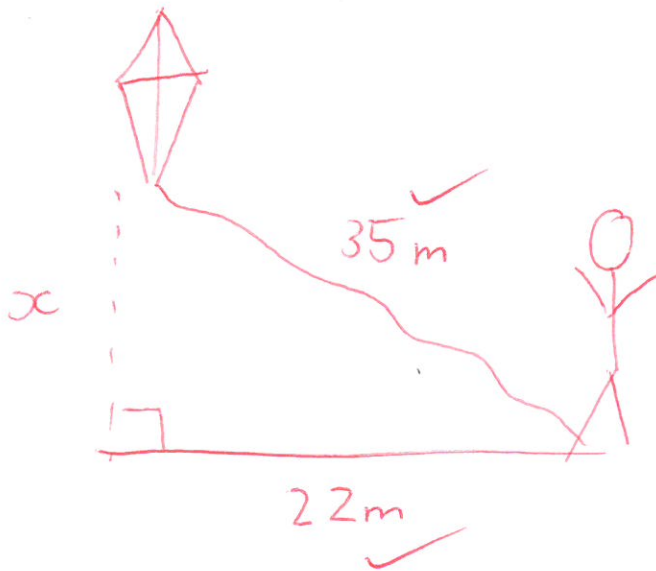
$$\sqrt{290.72} = 17.05 \text{ cm} \quad \checkmark$$



$$8^2 + 15^2 = 289 \quad \checkmark$$

$$\sqrt{289} = 17 \text{ m} \quad \checkmark$$

C. A kite is on a 35m string and is a horizontal distance of 22m from the person flying the kite. How high is the kite above the ground? Construct a diagram showing all the relevant details.



$$35^2 - 22^2 = 741 \quad \checkmark$$

$$\sqrt{741} \approx 27.22 \text{ m} \quad \checkmark$$